Lifeline

A standing safety policy for MUSD troves on Mezo

Set a rule once: "below 130%, put my reserve in until I'm back at 160%,
and keep 25,000 MUSD of other debt ahead of me in the redemption queue".
Anyone can enforce it. No one can abuse it.

Mezo Buildathon · Track 1 DeFi + Track 2 Access & Distribution

Two ways an MUSD borrower loses BTC

Liquidation

- Under **110% ICR**, or under **150%** in Recovery Mode
- BTC falls 20–30% in days
- Today's defense is to top up by hand at 3 AM

Redemption

- MUSD is swapped for BTC from the **lowest-ICR troves ≥ 110%** first
- Live testnet today: the front of the queue is at **114–131%**
- At 127% ICR, only **~18.6k MUSD** of debt stands in front of you

Fear of both keeps BTC holders from borrowing MUSD at all.

The solution

LifelineVault: a smart account that is the borrower of record for one MUSD trove.

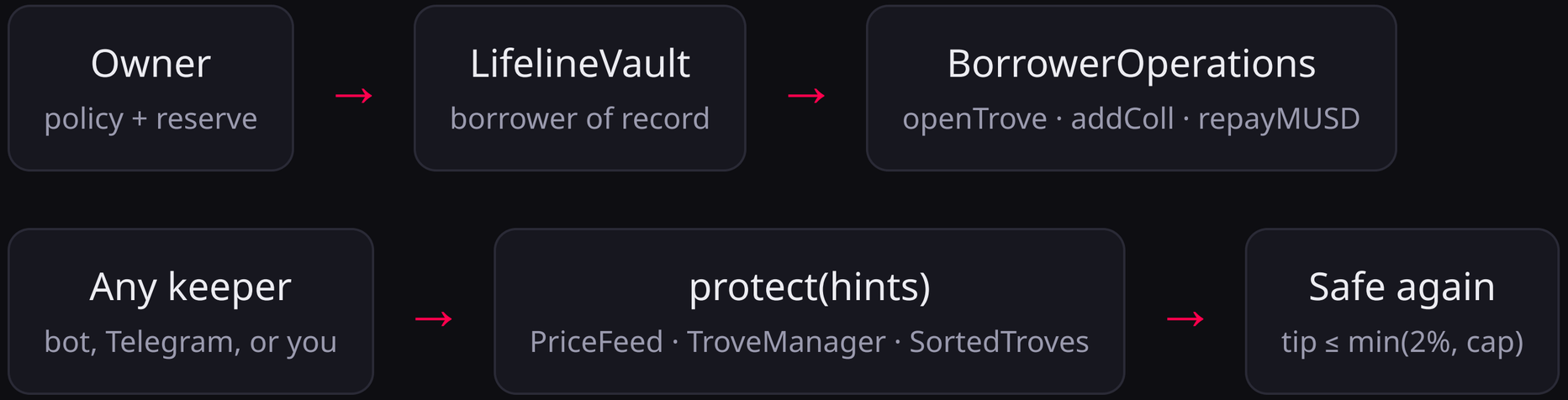
The owner parks a small **BTC and/or MUSD rescue reserve** and sets a policy: trigger · target · keeper tip (≤ 2%, capped) · redemption buffer

protect() is permissionless. It fires when ICR < trigger **or** the trove is inside the redemption buffer:

- adds BTC (`addColl`) or repays (`repayMUSD`)
- lifts the trove to safety, again and again
- pays the caller a capped tip

Keepers can only move the reserve **into the owner's own trove**. Withdrawals, borrowing and closing stay owner-only. Works for smart accounts; no one-shot pre-signed signatures.

How it plugs into mUSD



- **Recovery-Mode aware**: the trigger rises to ≥155% when system TCR < 150%
- **Exact HintHelpers hints**: opening on the 280-trove list costs ~0.66M gas instead of 6.3M
- A rescue that can't clear the liquidation line reverts instead of wasting funds

Demo 1: a 45% BTC crash on real mUSD contracts

```
ICR 190.7%  (trigger 130.0%, target 160.0%)  ok      liquidation price $48,444

== BTC crashes 45%: $84,000 -> $46,200. Both troves fall to ~105%
RESCUE 0x81c6…485F: ICR 104.9% < trigger 130.0%
Lifeline rescued: ICR 104.9% -> 160.0%, added 0.02626 BTC, keeper tip 0.00013 BTC

== A liquidator tries both troves
Bob (no Lifeline):  LIQUIDATED. His 0.05 BTC collateral is gone.
Alice (guarded):    liquidation REVERTED. Trove still active at the 160% target.
```

./demo/local-demo.sh forks Mezo testnet. Only the oracle precompile is mocked, so the price can move.

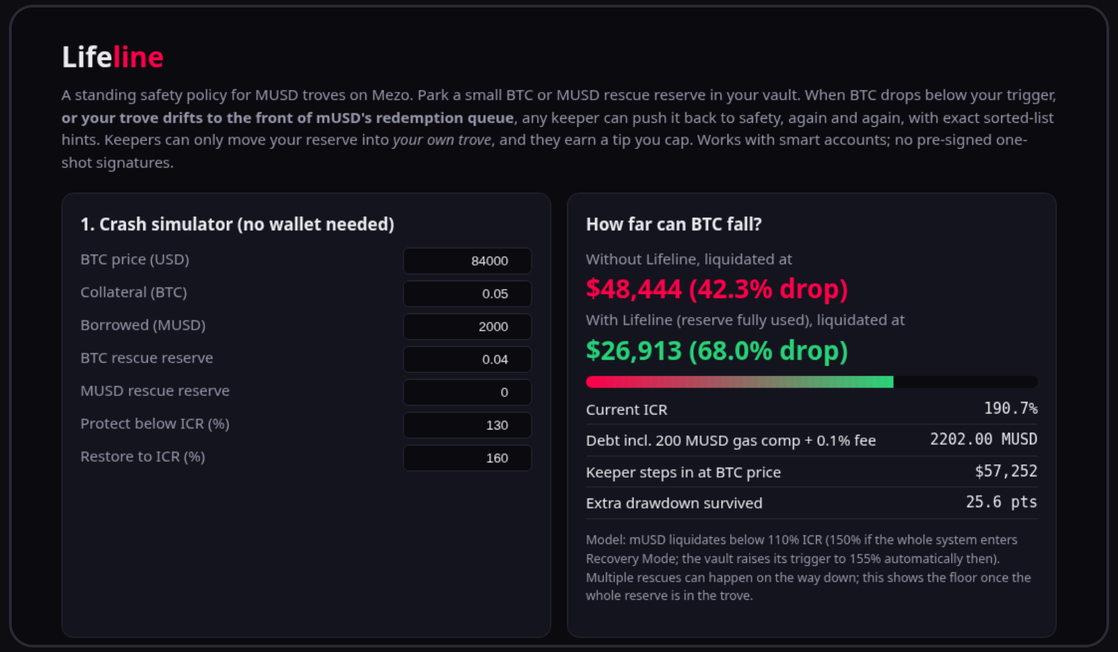
Demo 2: the redemption guard

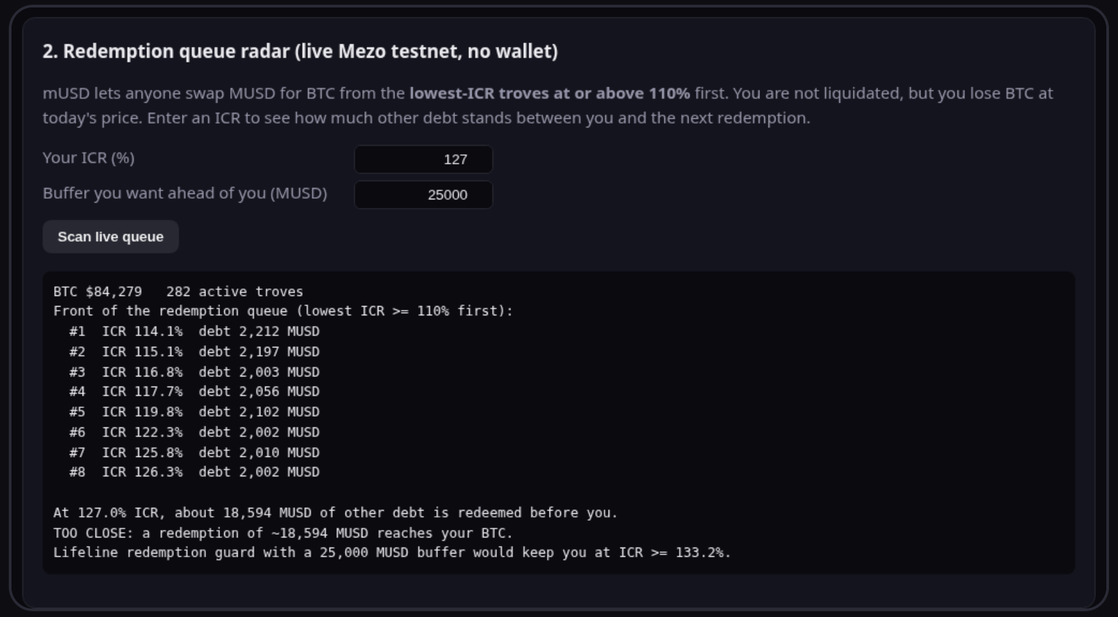
```
Carol opens at ~127% ICR: safe from liquidation, but near the FRONT of the redemption queue
RESCUE 0x8e9b…c965: front of the redemption queue (ICR 127.4% < safe 132.8%)
Lifeline rescued (redemption guard): ICR 127.4% -> 133.8%, added 0.00167 BTC

== A redeemer swaps 20,000 MUSD for BTC through mUSD's redemption mechanism
status 1 (success)
Carol (guarded):  collateral unchanged at 0.03507 BTC. The redemption never reached her.
```

Control test: **without** the guard, the same 20,000 MUSD redemption against the live trove list **takes Carol's BTC**.

Dashboard: crash simulator + live redemption radar





Same trove, same 0.04 BTC reserve: the liquidation price moves from **$48,444** to **$26,913**. The radar reads the live testnet queue with no wallet needed. Wallet actions: create vault → open trove with exact hints → watch health.

Live on Mezo testnet: real rescues, real txs

```
LifelineFactory  0x37aC7acd4e48b79bb0dc6AaCA5AF3611cF176149
Demo vault       0xC92BfDC83b03d32e39930Bfb789233e1Cc0fdbEF
Source verified on Sourcify (exact match)

open trove at 127.0% ICR (exact hints)                  tx 0xd624a32c...9da0
redemption-guard lift: front of queue, 127.0% -> 134.1% tx 0x2374b4b3...6e91
policy tightened to trigger 140% / target 150%          tx 0xeb93cddc...bdec
trigger rescue by keeper: 134.1% -> 150.0%              tx 0x1470fdfb...13ea
```

Real oracle, real 280-trove list, permissionless keeper. Links in the README and on **explorer.test.mezo.org**

The vault is healthy at 150% today: liquidation price ~$61.7k, redemption guard ok.

Why Lifeline

	Alerts only	One-shot pre-signed rescue	**Lifeline**
Acts while you sleep	no	once	every time
Rescue asset	–	repay MUSD	add BTC and/or repay
Recovery Mode (150%)	warn	no	trigger auto-raised
Redemption queue	monitor	monitor	on-chain guard
Smart accounts / Passport	–	EOA signature	any owner
Fork tests vs live mUSD	–	–	**16 / 16**

No admin, no upgrade keys, reentrancy guarded, double-capped tip, MIT, CI on every push. Not audited yet: testnet first.

Why it grows MUSD

- **More borrowing**: holders who fear liquidation *and* redemption can borrow with a standing safety net
- **Distribution**: an open-source keeper with Telegram alerts meets users where they already are (Track 2)
- **Composable**: built on the deployed mUSD contracts, with no protocol changes
- **Revenue path**: an optional, owner-chosen service fee on rescues; keepers earn tips
- **Wave 2**: auto-deleverage through a Mezo pool (no idle reserve), MUSD savings yield on the reserve, one-click migration of existing troves, a hosted alert bot, then mainnet after review

Lifeline

Never get liquidated, or redeemed, in your sleep.

Code, tests, keeper and demo: **github.com/XachP/mezo-lifeline**
Live dashboard: **xachp.github.io/mezo-lifeline**
Mezo testnet factory: **0x37aC7acd4e48b79bb0dc6AaCA5AF3611cF176149**

Built with an AI coding agent and reviewed by the submitter. MIT licensed.